

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

<think>

- First convert 1 eV to joules , knowing $1\text{ eV} \hat{=} 1.6 \times 10^{-19}\text{ J}$.
- Then use $E = hf$ and $\lambda = h/p$ where $p = \sqrt{2mK.E}$, to find λ .
- After calculation , the result $\hat{=} 5.27 \times 10^{-9}\text{ m}$ is matched most closely by option F.

</think>

<answer>F </answer>

